

Welcome to

Statistics and Probability Theory

VO 107.254 2026W

2 hours, 3 ECTS

UE 107.369 2026W

2 hours, 3 ECTS

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E105 – Institute of Statistics and Mathematical Methods in Economics

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The main goal of this course is to learn the basic concepts of Probability Theory (Stochastics) and Statistics, in two parts;

1. **PROBABILITY THEORY:** Theory of randomness
by Daniel Kapla
2. **STATISTICS:** Description of and inference from data
by Andrey Kofnov

Information and organization, online via;

TISS: Course announcement, description, *registration* and link to TUWEL.

TUWEL: Course *material + slides*, lecture *LiveStream + Recordings*, an *Introduction to R* and further course information.

The exercise sessions accompany the lecture by posting homework exercises covering the content of the lecture.

- Every week (starting today *October 5, 2026*) exercises are posted on TUWEL.
- The following week on Tuesdays, solutions are *presented by the students*.
- The *presentation is graded* by the tutor.
- Solved problems must be *ticked and uploaded in TUWEL* (till Monday 23:59).
- *At least 2 problems* solved *per attendance*.
- At most *2 absences* are allowed (no excuse required).
- At least *30 solved problems* in total.

Information and organization, online via;

TISS: Announcement, *registration to the course and a group* and link to TUWEL.

TUWEL: Detailed course information (Syllabus), additional material, exercise assignments and *instructions for submissions* as well as announcements.

LECTURES

Course Nr: 107.254

Time: *Monday*¹ from *8:00 – 10:00*

Place: Informatikhörsaal – ARCH-INF

First Class: In a few minutes.

Consultation: *Forum in TUWEL*
or by E-mail².

EXERCISES

Course Nr: 107.369

Time: *Tuesday*¹, see your registration

Place: See *your group* in the *Syllabus*

First Class: Next week, *October 13*

First Sheet: Today, *October 5*

Consultation: *Your tutor via E-mail*
see Syllabus in TUWEL.

¹Except on holidays, see Syllabus

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- Successful completion of *STEOP* (for informatics¹)
- The basics of *Linear Algebra* and *Calculus*.
- *Registration* in TISS *until 5.10.2026 23:59*
to the *lecture* and *exercise* courses and an exercise *group* (3 in total)
- For the *exercises*, a computer with the statistics software R;
R: Free and open source statistics software (required)
<https://cran.r-project.org>
RSTUDIO: Integrated Development Environment (IDE) for R (advised)
<https://posit.co/download/rstudio-desktop/>
- Prior knowledge of R is not needed.

¹Study Codes: 033 521, 033 526, 033 532, 033 533, and 033 534

EXAM: *One final* comprehensive *written exam* (based on course material)

- *20 Multiple-choice* questions.
- *Four choices*, exactly *one correct*.
- Testtime is *90 minutes*.
- You need a *non-programmable calculator*
- and a *handwritten two-sided A4 “cheat sheet”* is allowed.
- *Smartphones and -watches*, computers, tablets, notes, books, ... as well as discussions and/or consultations are *prohibited* during the exam!

DATES: Three examination dates are provided (at least);

1. Friday *29.1.2027* from *11:00 – 13:00*
2. Friday *26.2.2027* from *13:00 – 15:00*
3. Monday *5.4.2027* from *13:00 – 15:00*

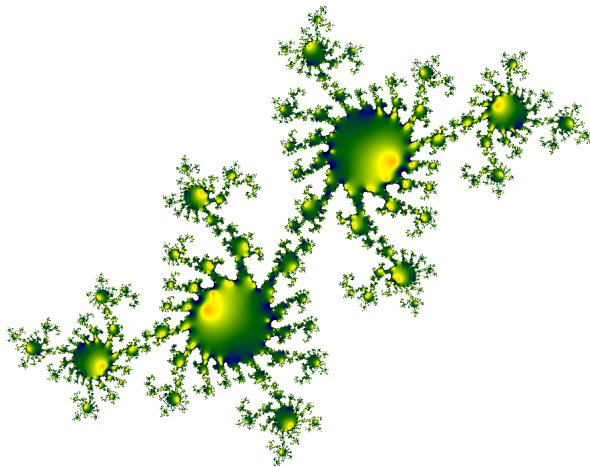
REGISTRATION: in TISS.

Note: *Students* who are *registered but do not attend the exam* will be *blocked* for registration to the next exam!

The main reference for the schedule, operations, grading, ... and alike for the VO and UE are the two *Syllabus* files.

See *Syllabus* in TISS and TUWEL!

If there are any errors, oversights, ambiguousness or obscurities,
please let us know!



Probability Theory vs. Statistics

Probability and Statistics are different subjects but both about *random processes*.

DETERMINISTIC

The outcome is *fixed* (we are *certain about the value of a variable*)

- Your name
- Date of easter
- The outcome of yesterday's game
- The surface gravity on Mercury

RANDOM

There are multiple outcomes (we are *uncertain about the value of a variable*)

- The name of the next person entering this room
- Date of your final exam
- The outcome of tomorrow's game
- The weather in a month

Probability Theory vs. Statistics

PROBABILITY: The *population* (the whole) is *known* and the probabilities about members of the population (subsets, events) are *computed*.

- Logical and self-contained
- Rules for calculating probabilities
- One correct answer

STATISTICS: *Members of the population* (data, events) are *observed* and we *infer* about the population (the whole.)

- Messier and more of an “art”
- Collecting data and observing events to draw probabilistic conclusions
- No single correct answer

PROBABILITY EXAMPLE: A *fair* coin is tossed 100 times. What is the probability of getting 60 or more heads?

1. There is *one answer*, which is $\approx 2.84\%$ (we'll learn how to compute it)
2. The *random process is known* (probability for heads is 50%)
3. We seek the *probability* of an *undetermined event* (at least 60 heads)

STATISTICS EXAMPLE: We have a *coin of unknown origin*. To investigate if the coin is fair, we toss it 100 times and count 60 heads. As statisticians, we have to draw a conclusion (inference) from this data.

1. There are *many ways*, different statistics can draw different conclusions
2. The *process is known*, only observation are (60 heads)
3. We seek to *explain the random process* (probability of heads)

Both have a wide range of applications, from the social to the natural sciences.

- Testing one medical treatment vs. another (or placebo)
- Optimization and Machine Learning (and AI)
- Gambling, betting, and strategies
- Insurance and Investment
- Computer-games and -simulations up to Computergraphics (rendering)
- Epidemiology
- Marketing
- ...

We are going to study *toy models*, like *coins*² and *dice*.

²A realistic model for all situations with 2 outcomes.

Intuitively one can say;

Probability is a *measure of uncertainty* of the occurrence *of an event*.

With an *event* being all *desired outcomes* out of all *possible outcomes*.

Mathematically³; the *Probability* $P(A)$ of an *event* A out of *all outcomes* Ω as

$$P(A) = \frac{\text{nr. of desired outcomes}}{\text{nr. of all outcomes}} = \frac{|A|}{|\Omega|}.$$

³For a finite sample space with equally likely outcomes. An assumption for now!

What is the *probability* of getting *exactly 1 head* out of *3 tosses*?

SAMPLE SPACE: Listing all 8 *possible outcomes*

$$\Omega = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$$

EVENT: There are 3 *outcomes* in the “*exactly 1 head*” event

$$A = \{HTT, THT, TTH\}$$

PROBABILITY: With all outcomes being equally likely; we get

$$P(A) = \frac{|A|}{|\Omega|} = \frac{3 \text{ outcomes with 1 head}}{8 \text{ possible outcomes}} = \frac{3}{8} = 37.5\%.$$

THE PRINCIPLE

Suppose there are n *possible outcomes*, each equally likely. For k *desired outcomes*, the *probability* of a desired outcome is k/n .